

Abstract

Pension Risk Transfer (PRT) transactions require insurers to assume large, long-dated annuity liabilities whose true risk profile is shaped by heterogeneous longevity dynamics, benefit concentration, and demographic structure. Traditional portfolio monitoring relies on aggregated reserve and demographic summaries, which obscure multivariate structural risks that emerge at the subpopulation level. This study develops a quantitative framework to identify and measure latent structural risk within PRT portfolios using unsupervised learning and actuarial weighting.

Using engineered demographic, health, occupational, benefit, and socioeconomic features, Principal Component Analysis (PCA) is applied to extract latent actuarial risk factors governing portfolio heterogeneity. k-means clustering on PCA scores segments the portfolio into structurally distinct annuitant subpopulations capturing differences in benefit scale, retirement timing, and socioeconomic-health profiles. These components are combined with longevity-weighted liability exposure and an entropy-based concentration penalty to construct a unified Portfolio Quality Score (PQS), designed to summarize the portfolio's structural risk in a single interpretable metric.

Applied to a synthetic portfolio of 20,000 annuitants, the model identifies three dominant latent factors explaining approximately 45% of total variance and reveals actuarially meaningful clusters reflecting concentrated benefit exposure and divergent longevity patterns. The resulting PQS of 2.047 indicates moderate structural risk driven primarily by longevity dispersion and benefit concentration. Sensitivity analysis demonstrates coherent responses to global and targeted longevity shocks, liability reallocations, and outlier removal. Model validation confirms high robustness, with cluster stability under bootstrap resampling yielding an Adjusted Rand Index of 0.982 and minimal PQS variability under perturbation and subsampling.

This research demonstrates that unsupervised learning can surface structural PRT risks invisible to traditional dashboards. The PQS framework provides a forward-looking, data-driven tool for portfolio monitoring, pricing diagnostics, and assumption governance, offering a practical blueprint for integrating machine learning into actuarial risk management.